

Generation Interconnection System Impact Study Report

For Queue Project Q20-1512

Bliss Windpark 100 kV

August 23, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1512. The Interconnection Customer proposes a 159.8 MW solar photovoltaic facility interconnecting at the Bliss Windpark 100 kV substation in the West Virginia area. The request was received on July 02, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 541 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,092,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Bliss Windpark 100 kV/1 kV Transformer Bank
Contingency: Bliss Windpark-Firstenergy W H Sammis 100kV
Loading: 226.3% (Rating: 71 MVA)
DFAX: 100.0%
Pre-project loading: 0.0%
2. Facility: Bliss Windpark-Firstenergy W H ammis 100kV line
Contingency: Bliss Windpark 100 kV/1 kV TX
Loading: 126.5% (Rating: 126 MVA)
DFAX: 100.0%
Pre-project loading: 0.0%
3. Facility: New Albany Grant Line - He Substation-Firstenergy W H Sammis 100kV L
Contingency: Holt Switch-Eighteenth St 100kV
Loading: 125.3% (Rating: 131 MVA)
DFAX: 49.9%
Pre-project loading: 64.4%
4. Facility: New Albany Grant Line - He-Lerdo 100kV
Contingency: Holt Switch-Eighteenth Street 100kV
Loading: 111.7% (Rating: 136 MVA)
DFAX: 49.9%
Pre-project loading: 53.3%
5. Facility: Holt Switch-Eighteenth Street 100kV
Contingency: New Albany Grant Line - He Substation-Firstenergy W H ammis 100kV
Loading: 109.4% (Rating: 130 MVA)
DFAX: 28.3%
Pre-project loading: 74.7%
6. Facility: Bliss Windpark 138/1 kV TX
Contingency: Firstenergy W H Sammis-Bliss Windpark 100kV
Loading: 100.0% (Rating: 154 MVA)

DFAX: 100.0%
Pre-project loading: 4.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bliss Windpark 100 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$3,047,000	100% (\$3,047,000)
Bliss Windpark-Firstenergy W H Sammis Substation 100kV L	Rebuild line with higher-capacity conductor	\$2,208,000	100% (\$2,208,000)
New Albany Grant Line - He Substation-Firstenergy W H Sammis 100kV	Rebuild line with higher-capacity conductor	\$3,603,000	40% (\$1,448,000)
New Albany Grant Line - He-Lerdo 100kV	Rebuild line with higher-capacity conductor	\$2,174,000	40% (\$873,000)
Eighteenth St-Holt Switch 100kV	Reconductor line	\$3,395,000	27% (\$905,000)
Bliss Windpark 138/1 kV Auto	Replace transformer with higher-rated unit	\$4,611,000	100% (\$4,611,000)
	Total allocated to project		\$13,092,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.